Term Project

Class: CIS 4004

Semester: Spring 2026

Due Dates: Monday, April 5

Term Project: a MERN-Stack Web Application

With this assignment, students in groups of 5 will create a web application using the MERN stack, around a theme of their choice. Students will apply their knowledge of front-end and back-end development using Node.js, Express.js, MongoDB, and React. The learning outcomes for this assignment are as follows:

- Students will be able to develop a Single-Page Application, using React.
- Students will be able to develop a web server that uses routing, using Express.js on Node.js.
- Students will be able to facilitate CRUD operations for MongoDB.
- Students will be able to combine front-end and back-end development to create a full-stack solution that allows users to perform CRUD operations through the single-page application.

Students are expected to work as a team to accomplish this project. **Every student is expected to substantially contribute to the solution**. All team members must be present for the project presentation (to be scheduled).

Submission Requirements

Each group will develop a full-stack solution using MERN technologies. **Do not deviate from this stack**. Students are not required to deploy the solution to a server. Instead, the project deliverables will be packaged in a .zip folder and submitted to WebCourses. This will include the front-end solution and the back-end solution, preferably in marked directories. Groups may choose any theme around which the solution must be based; this can be a hobby, interest, fictional business, etc. Please choose an appropriate theme. While the theme is open to selection, the project **must** meet the following requirements:

1. The web application must be developed using MERN: Node.js, Express.js, MongoDB, React.
2. There should be at least two types of users: Administrator and Standard User. Users must login to the application.
3. The first screen shown by the application should be the login page; new users should be able to be created by taking in a username and password. No duplicate usernames should be allowed.
4. Subsequent screens should be shown to the correct users; for example, a customer should not see a screen to manage inventory, because that is an Administrator task.
5. Administrators should be given a separate list of functionalities than Standard Users.
6. Administrators must be able to perform all CRUD operations in some capacity (Create, Read, Update, and Delete).
7. Standard Users must be able to read data that makes sense for them to see, and they must be able to create / update / delete their own data. A standard user should not be permitted to update or delete another user's data.
8. The data model must include at least 5 entities for which data will be stored.
9. There should be at least one "Many-to-Many" relationship in the data model.

Presenting your Project

Every group will be randomly assigned a date for their presentation. Each group will be allocated **10 minutes** to present their work. The podium has an HDMI cable, and groups will present their project to the rest of the class on their own computer. Presentations must follow this format:

- Introduction
  - Who is on your team?
  - What is your topic?
  - Why did you choose this topic?
  - What is the relevant information for the topic that makes it interesting?
- Distribution of Work
  - What technologies were leveraged for the project?
  - Who did what for the project?
- Technical Description
  - What form did routing take, for the web server?
  - What form did the collections take, for the database?
  - What form did the components take, for the front-end application?
  - What entities existed in the data model?
  - What many-to-many relationship(s) existed in the data?
- Live Demonstration

Use of PowerPoint slides (or equivalent) is encouraged. Ideally, the slides will be finished in 5 minutes, and the live demonstration will take 5 minutes.

Submitting Your Project

1. In WebCourses, navigate to the assignment and upload a single .zip file containing your entire submission. Only one member of the group needs to upload the project. **Please coordinate to make sure it gets submitted**. **Be careful to include everything needed to run your application** – the Node/Express project as well as the React application project!
2. In addition, please include **documentation** for your project, inside a "README.txt" file:
    a. How can the grader start your web server?
    b. Is there a second server needed for your React application? If so, explain how to start.
    c. How can the grader navigate to the application? I.e., what port and/or URI?
    d. What Collections are needed in MongoDB?
3. Lastly, **every** student will submit a reflection that describes the following:
    a. What did you accomplish? Be specific.
    b. What did your teammates accomplish? Be specific.
    c. What challenges did you encounter while working on the project, and how did you overcome them?
    d. If you had more time, what else would you do to improve the submission?

Grading Your Assignment

Your assignment will be scored based on multiple factors, described below. Your grade will be posted to WebCourses. If you have any questions about your grade, please contact the instructor. Any concerns about grades must be resolved within 1 week of the grade being posted.

Grading Criteria:

If the solution does not use the MERN stack, the submission will be given a 0.

- **(50%) Required Functionality** – the project will be scored, in part, based on the submission's adherence to the required functionality as described above.
- **(20%) Visually Appealing Design** – the project will be scored, in part, based on the submission's design. Students have the choice to write custom CSS or to leverage an online framework such as Bootstrap, or a combination of the two.
- **(20%) Presentation** – the project will be scored, in part, based on the group presentation's adherence to the above criteria. The presentation must be delivered in a professional manner. The presentation must fall within the time constraints.
- **(10%) Reflection** – the project will be scored, in part, based on the student reflection's adherence to the above criteria.

An improper distribution of work will result in a penalty to the grade. In other words, **every student must contribute to this project meaningfully**.